Regularización: Ridge, Lasso y Elastic Net

Big Data y Machine Learning para Economía Aplicada

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Agenda

1. Previously...
   - The objective
   - Regularization
   - Ridge
2. Lasso
   - Ridge and Lasso: Pros and Cons
3. Familia de regresiones penalizadas
4. Elastic Net

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El objetivo

- Para cualquier variable aleatoria Y y vector de covariables W, tenemos:

$$y = m(W) + u$$

- Bajo pérdida cuadrática, para cualquier función medible $m(W)$:

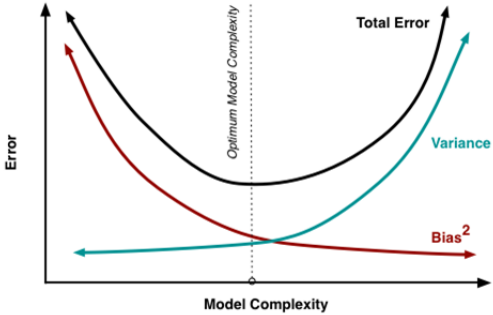
$$\mathbb{E}[Y \mid W] \;=\; \arg\min_{m} \; \mathbb{E}\left[(Y - m(W))^2\right]$$

- El problema práctico
  - La CEF es el objetivo, pero:
    1. No observamos W directamente — observamos X (aproximación ruidosa).
    2. Desconocemos la forma funcional de $m(W)$.

Error de predicción fuera de muestra

- ML nos interesa la predicción fuera de muestra
- Para un nuevo punto x_0:

$$\text{Error de Predicción Esperado}(x_0) = \text{Sesgo}^2 + \text{Varianza} + \underbrace{\sigma_u^2}_{\text{irreducible}} \tag{1}$$



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Regularization

- Asegurar cero sesgo dentro de muestra crea problemas fuera de muestra: trade-off Sesgo-Varianza
- Las técnicas de machine learning fueron desarrolladas para hacer este trade-off de forma empírica.
- Vamos a proponer modelos del estilo

$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - \beta_0 - x_{i1}\beta_1 - \cdots - x_{ip}\beta_p)^2 + \lambda \sum_{j=1}^{p} R(\beta_j) \tag{2}$$

- donde R es un regularizador que penaliza funciones que crean varianza

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Ridge

- Para un $\lambda \geq 0$ dado, consideremos ahora el siguiente problema de optimización

$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - \beta_0 - x_{i1}\beta_1 - \cdots - x_{ip}\beta_p)^2 + \lambda \sum_{j=1}^{p} (\beta_j)^2 \tag{3}$$

Example

photo from https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/

Ridge as Data Augmentation

RidgeDataAug

- Add λ additional points

$$\sum_{i=1}^{n}(y_i - x_i\beta)^2 + \lambda\beta^2 \tag{4}$$

Ridge vs OLS

- Ridge es sesgado $E(\hat{\beta}_{ridge}) \neq \beta$

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- Pero la varianza es menor que la de OLS

Ridge vs OLS

- Ridge es sesgado $E(\hat{\beta}_{ridge}) \neq \beta$
- Pero la varianza es menor que la de OLS
- Para ciertos valores del parámetro $\lambda \Rightarrow MSE_{OLS} > MSE_{ridge}$

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Lasso

- Para un $\lambda \geq 0$ dado, consideremos el siguiente problema de optimización

$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - \beta_0 - x_{i1}\beta_1 - \cdots - x_{ip}\beta_p)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \tag{5}$$

Lasso

- Para un $\lambda \geq 0$ dado, consideremos el siguiente problema de optimización

$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - \beta_0 - x_{i1}\beta_1 - \cdots - x_{ip}\beta_p)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \tag{5}$$

  - "LASSO's free lunch": selecciona automáticamente los predictores que van en el modelo ($\beta_j \neq 0$) y los que no ($\beta_j = 0$)
  - Por qué? Los coeficientes que no van son soluciones de esquina
  - $L(\beta)$ es no differentiable

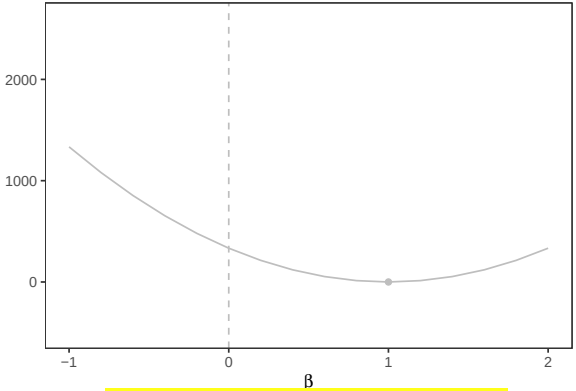
Intuición en 1 Dimensión

$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 + \lambda|\beta| \tag{6}$$

Intuición en 1 Dimensión

$\hat{\beta} > 0$

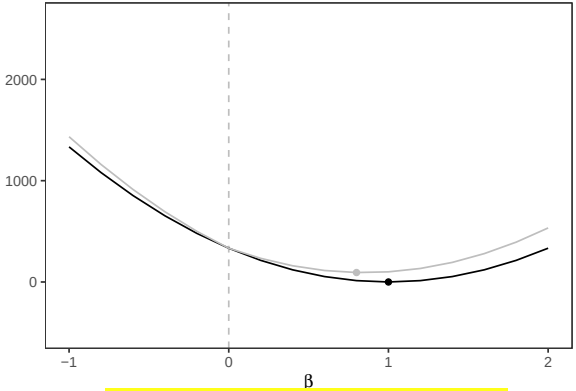
$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 + \lambda\beta \tag{7}$$



Intuición en 1 Dimensión

$\hat{\beta} > 0$

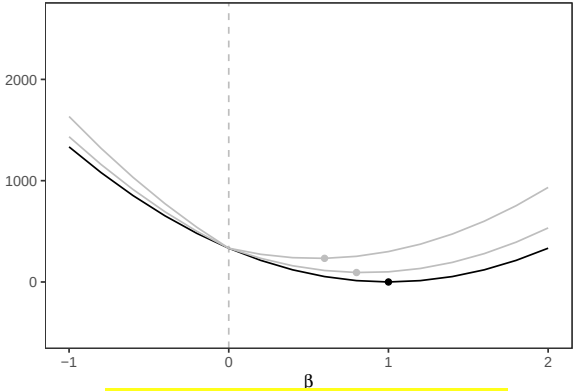
$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 + \lambda\beta \tag{8}$$



Intuición en 1 Dimensión

$\hat{\beta} > 0$

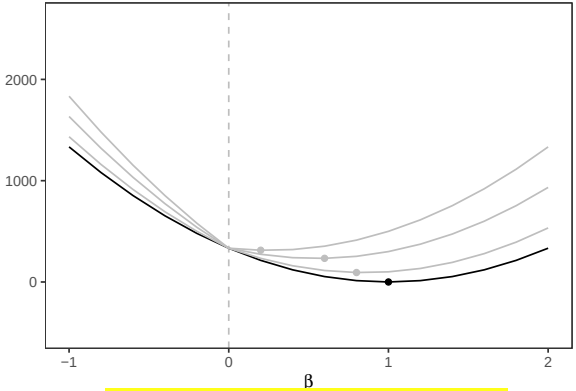
$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 + \lambda\beta \tag{9}$$



Intuición en 1 Dimensión

$\hat{\beta} > 0$

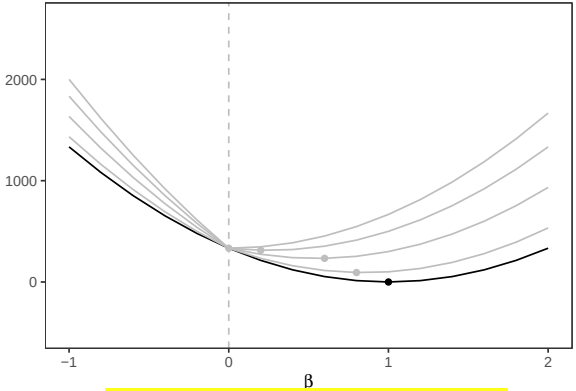
$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 + \lambda\beta \tag{10}$$



Intuición en 1 Dimensión

$\hat{\beta} > 0$

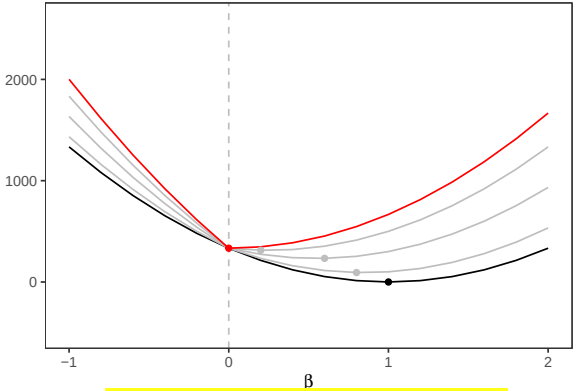
$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 + \lambda\beta \tag{11}$$



Intuición en 1 Dimensión

$\hat{\beta} > 0$

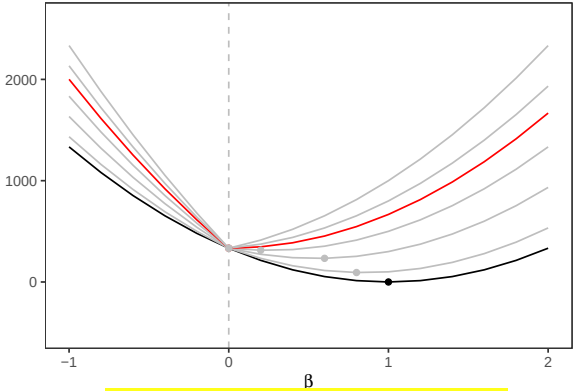
$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 + \lambda\beta \tag{12}$$



Intuición en 1 Dimensión

$\hat{\beta} > 0$

$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 + \lambda\beta \tag{13}$$



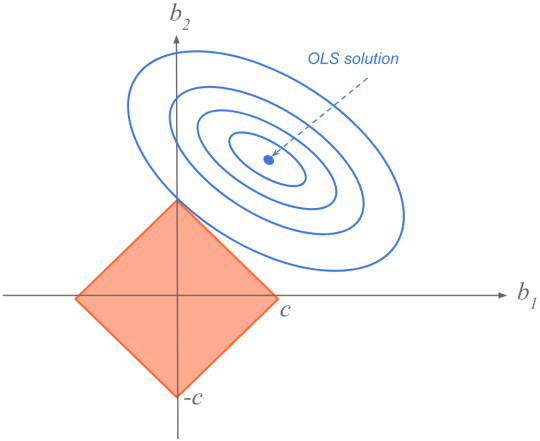
Intuición en 1 Dimension

Solución analitica

$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - x_i\beta)^2 + \lambda|\beta| \tag{14}$$

Intuición en 2 Dimensiones (Lasso)

$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - x_{i1}\beta_1 - x_{i1}\beta_2)^2 \text{ s.a } (|\beta_1| + |\beta_2|) \leq c \tag{15}$$



Example

photo from https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/

Resumen

- Ridge y Lasso son sesgados, pero las disminuciones en varianza pueden compensar estoy y llevar a un MSE menor
- Lasso encoje a cero, Ridge no tanto
- Importante para aplicación:
  - Estandarizar los datos
  - Elegimos $\lambda \rightarrow$ Validación cruzada

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Ridge and Lasso: The good and the bad

- Objective 1: Accuracy
  - Minimize prediction error (in one step) $\rightarrow$ Ridge, Lasso
- Objective 2: Dimensionality
  - Reduce the predictor space $\rightarrow$ Lasso's free lunch
- More predictors than observations ($k > n$)
  - OLS fails
  - Ridge augments data
  - Lasso chooses at most n variables

Ridge and Lasso: The good and the bad

OLS when $k > n$

- Rank? Max number of rows or columns that are linearly independent
  - Implies $rank(X_{n\times k}) \leq min(k, n)$
- MCO we need $rank(X_{n\times k}) = k \implies k \leq n$
- If $rank(X_{n\times k}) = k$ then $rank(X'X) = k$
- If $k > n$, then $rank(X'X) \leq n < k$ then $(X'X)$ cannot be inverted
- Ridge works when $k \geq n$

Ridge and Lasso: The good and the bad

Ridge when $k > n$

$$min_{\beta} E(\beta) = \sum_{i=1}^{n} (y_i - \sum_{j=1}^{k} x'_{ij}\beta_j)^2 + \lambda(\sum_{j=1}^{k} \beta_j)^2 \tag{16}$$

- Solution $\rightarrow$ data augmentation
- Intuition: Ridge "adds" k additional points.
- Allows us to "deal" with $k \geq n$

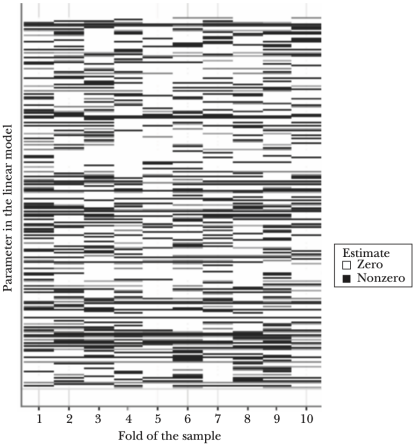
Ridge and Lasso: The good and the bad

Ridge when $k > n$

Ridge and Lasso: The good and the bad

- When we have a group of highly correlated variables,
  - Lasso chooses only one.

Ridge and Lasso: The good and the bad



Ridge and Lasso: The good and the bad

- When we have a group of highly correlated variables,
  - Lasso chooses only one. Makes it unstable for prediction.
  - Ridge shrinks the coefficients of correlated variables toward each other. This makes Ridge "work" better than Lasso. "Work" in terms of prediction error

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Family of penalized regressions

$$min_{\beta} R(\beta) = \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \sum_{s=2}^{p} |\beta_s|^q \tag{17}$$

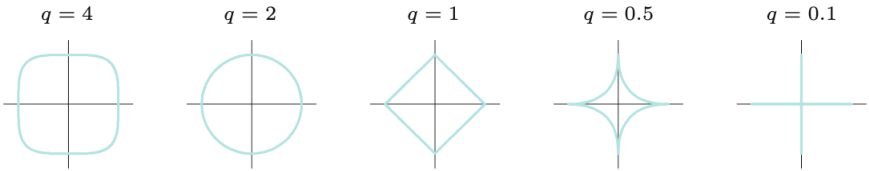

FIGURE 3.12. *Contours of constant value of* $\sum_j |\beta_j|^q$ *for given values of* q.

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Elastic net

$$min_{\beta} EN(\beta) = \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j)^2 + \lambda \left(\alpha \sum_{j=1}^{p} |\beta_j| + \frac{(1-\alpha)}{2} \sum_{j=1}^{p} (\beta_j)^2 \right) \tag{18}$$

- Si $\alpha = 1$ Lasso
- Si $\alpha = 0$ Ridge

Elastic Net

- Elastic net: happy medium.
  - Good job at prediction and selecting variables

$$min_{\beta} EN(\beta) = \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j)^2 + \lambda \left(\alpha \sum_{j=1}^{p} |\beta_j| + \frac{(1-\alpha)}{2} \sum_{j=1}^{p} (\beta_j)^2 \right) \tag{19}$$

- Mixes Ridge and Lasso
- Lasso selects predictors
- Strict convexity part of the penalty (ridge) solves the grouping instability problem
- How to choose (λ, α)? $\rightarrow$ Bidimensional Crossvalidation
- Recomended lecture: Zou, H. & Hastie, T. (2005)

Example

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